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ESTIMATION PRO — FIXED RATE BAND COST ESTIMATE

Project: Quick Analysis

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CALCULATION MANIFEST — SINGLE SOURCE OF TRUTH

--- All locked input parameters and cost outputs for this estimate. Sections 1 and 7 copy values from this table verbatim without re-derivation.

Parameter	Value
Fabricated Tonnage — AI Extracted	4.92 t
Material Type	Carbon Steel
Surface Treatment	Shop primer only
Assembly Complexity	Mixed bolt and weld
Material Factor	x1.00
Surface Factor	x1.00
Assembly Factor	x1.10
Composite Adjustment Factor	x1.10
Rate Basis	DEFAULT
Rate — Low (USD per ton)	$3,500
Rate — Mid (USD per ton)	$4,000
Rate — High (USD per ton)	$4,500
Adjusted Rate — Low	$3,850

Parameter	Value
Adjusted Rate — Mid	$4,400
Adjusted Rate — High	$4,950
Country	USA
Tax Rate	0%
LOCK COST — LOW	$18,900
LOCK COST — MID	$21,600
LOCK COST — HIGH	$24,400
Cost per Fabricated Ton — Mid	$4,390

All values above are locked for this document. Sections 1 and 7 reproduce these values exactly. No recalculation permitted.

1. EXECUTIVE SUMMARY

--- Fabrication cost summary for this project at the applied rate band.

Metric	Value
Project Name or ID	HOME 2 - CAROWINDS
Country	USA
Total Fabricated Tonnage	4.92 t
Material Type	Carbon Steel
Surface Treatment	Shop primer only
Assembly Complexity	Mixed bolt and weld
Composite Adjustment Factor	x1.10
Rate Basis	DEFAULT

Metric	Value
Rate Band Applied	$3,500 — $4,500 per fabricated ton
Tax Rate	0%
Estimated Cost — Low (inc. tax)	$18,900
Estimated Cost — Mid (inc. tax)	$21,600
Estimated Cost — High (inc. tax)	$24,400
Cost per Fabricated Ton — Mid	$4,390 per ton

2. BASIS OF ESTIMATE

--- Drawings reviewed: S0.01 GENERAL NOTES S0.02 TYPICAL SCHEDULES & DETAILS S0.03 SPECIAL INSPECTIONS S1.01 FOUNDATION PLAN S1.02 SECOND FLOOR FRAMING PLAN S1.03 TYPICAL LEVEL FRAMING PLAN S1.04 ROOF FRAMING PLAN S2.01 STRUCTURAL WALL DESIGNATION PLAN S2.02 CMU AND WOOD STUD SCHEDULES S2.03 CMU REINFORCEMENT ELEVATIONS - STAIR #1/ELEVATOR S2.04 CMU REINFORCEMENT ELEVATIONS - STAIR #2 S3.01 FOUNDATION SECTIONS S3.02 FLOOR FRAMING SECTIONS S3.03 ROOF FRAMING SECTIONS S3.04 ROOF FRAMING SECTIONS S3.05 CANOPY FRAMING SECTIONS S3.06 TRELLIS AND SITE WALL DETAILS

Tonnage extraction method: Method C — Estimated from framing plans where a BOM is not present. Member quantities and lengths were taken from sheets S1.02 and S3.06.

Composite factor selection: Material factor x1.00: Carbon Steel (ASTM A992, A500, A36) specified in General Structural Notes on sheet S0.01. Surface factor x1.00: "ALL EXPOSED STRUCTURAL STEEL SHALL BE SHOP PAINTED WITH ONE COAT OF PRIMER" specified in General Structural Notes on sheet S0.01. Assembly factor x1.10: Determined from the scope of work visible on the drawings, which includes both welded components (lintel plates on S0.02) and bolted connections (detail 1/S0.02).

Rate basis: Rate band of $3,500 to $4,500 per fabricated ton applied. Midpoint of $4,000 per ton is the headline figure.

Tax: 0% applied for USA.

Scope basis: The tonnage figure includes W-shapes, HSS members, and plates that are explicitly shown on the Second Floor Framing Plan (S1.02) and Trellis Details (S3.06) and for which a unit weight is available in the AISC reference table. Excluded from this estimate are all lintels (W8x24, W16x50, L7x4x3/8, L6x4x3/8, L4x4x3/8) and columns (C5, C5A, C6, C8B) from the schedules on S0.02 as their quantities and locations were not specified on the framing plans. Also excluded are all HSS members whose profiles (HSS18x6x3/8, HSS9x5x3/8, HSS8x6x3/8) do not appear in the provided AISC Unit Weight

Reference table.

3. MEMBER-BY-MEMBER TONNAGE TAKE-OFF

--- Table 3A — Primary Structural Members All columns, primary beams, moment frames, transfer members, and lateral bracing elements extracted from the drawing package.

No.	Type	Mark	Profile	Qty	Length Imperial	Length mm	Unit Wt kg/m	Est Wt kg	Est Wt lb	Grade	Source Sheet	Method
1	BEAM	—	W16x36	1	16'-11"	5,156	53.6	276	609	A992	S1.02	PLAN ESTIMATE
2	BEAM	—	HSS18x6x3/8	3	—	—	NOT IN TABLE	NOT IN TABLE	NOT IN TABLE	A500	S1.02	PLAN ESTIMATE
3	BEAM	—	HSS8x6x3/8	2	—	—	NOT IN TABLE	NOT IN TABLE	NOT IN TABLE	A500	S3.06	PLAN ESTIMATE
4	COLUMN	—	HSS8x6x3/8	4	—	—	NOT IN TABLE	NOT IN TABLE	NOT IN TABLE	A500	S3.06	PLAN ESTIMATE

Table 3B — Secondary and Miscellaneous Members All secondary beams, purlins, girts, joists, bracing, stairs, handrails, platforms, and miscellaneous structural steel from drawings.

No.	Type	Mark	Profile	Qty	Length Imperial	Length mm	Unit Wt kg/m	Est Wt kg	Est Wt lb	Grade	Source Sheet	Method
1	BEAM	—	W14x22	34	12'-10"	3,912	32.7	4,347	9,584	A992	S1.02	PLAN ESTIMATE
2	HEADER	—	HSS9x5x3/8	1	—	—	NOT IN TABLE	NOT IN TABLE	NOT IN TABLE	A500	S1.02	PLAN ESTIMATE
3	BEAM	—	HSS6x4x1/4	1	4'-0"	1,219	18.0	22	48	A500	S1.02	PLAN ESTIMATE

Table 3C — Connection Material and Plates All shear plates, base plates, stiffeners, gusset plates, and embedded plates. Show the full plate formula calculation per row.

No.	Description	Thickness mm	Width mm	Length m	Qty	Wt kg	Wt lb	Source Sheet	Method
1	Trellis Base Plate	25.4	406	0.406	4	131	290	S3.06	PLAN ESTIMATE

Table 3D — Tonnage Subtotals by Category Weight consolidated by member category. The TOTAL row must equal LOCK_TONNAGE including the 3.00% accessory addition.

Category	Count	Total Wt kg	Total Wt lb	Metric Tons
W-Shapes — Columns and Beams	35	4,623	10,193	4.62
HSS and Tube	10	22	48	0.02
Pipe	0	0	0	0.00

Category	Count	Total Wt kg	Total Wt lb	Metric Tons
Angles and Channels	0	0	0	0.00
Plates — all types	4	131	290	0.13
Miscellaneous and Embeds	0	0	0	0.00
Bolts and Accessories 3.00%	—	144	318	0.14
TOTAL	49	4,920	10,848	4.92

4. COST BUILD-UP BY MEMBER CATEGORY

--- Fabrication cost for each category at all three rate points. Adjusted rate = base rate x composite factor applied uniformly to all categories.

Category	Tons	Adj Rate Low	Adj Rate Mid	Adj Rate High	Cost Low USD	Cost Mid USD	Cost High USD	Pct
W-Shapes	4.62	$3,850	$4,400	$4,950	$17,787	$20,328	$22,869	94%
HSS and Tube	0.02	$3,850	$4,400	$4,950	$77	$88	$99	0%
Pipe	0.00	$3,850	$4,400	$4,950	$0	$0	$0	0%
Angles and Channels	0.00	$3,850	$4,400	$4,950	$0	$0	$0	0%
Plates	0.13	$3,850	$4,400	$4,950	$501	$572	$644	3%
Miscellaneous and Embeds	0.00	$3,850	$4,400	$4,950	$0	$0	$0	0%

Category	Tons	Adj Rate Low	Adj Rate Mid	Adj Rate High	Cost Low USD	Cost Mid USD	Cost High USD	Pct
Bolts and Accessories	0.14	$3,850	$4,400	$4,950	$539	$616	$693	3%
TOTAL	4.92	$3,850	$4,400	$4,950	$18,904	$21,604	$24,305	100%

5. PROCESS COST BREAKDOWN — MIDPOINT SCENARIO

--- $21,600 distributed across fabrication process activities at fixed industry-standard percentage allocations.

Process Activity	Industry Pct	Amount USD	Notes
Material — mill sections and plate	35%	$7,600	Structural steel mill cost
Shop labour — cut, fit, and weld	30%	$6,500	AWS D1.1 weld operations included
Coatings and surface preparation	12%	$2,600	SSPC class as specified
Consumables — weld wire, bolts, gas	6%	$1,300	Per-ton standard allowance
Equipment — plasma, burn table, drill	7%	$1,500	CNC and NC machine time
Overhead and QA/QC	6%	$1,300	Inspection, NDT, project management
Shop margin	4%	$800	Fabricator margin
TOTAL	100%	$21,600	Midpoint scenario

6. COST WATERFALL

--- Sequential cost build from LOCK_TONNAGE to final Grand Total at each rate point showing composite adjustment and tax.

Step	Calculation Detail	Amount USD
Fabricated tonnage	4.92 t	—
Adjusted mid rate	$4,000 x 1.10	$4,400 per ton
Base cost at mid rate	4.92 t x $4,400	$21,648
Tax at 0%	Base cost x 0%	+$0
Grand Total — Low ($3,500 base x composite x tax)	—	$18,900
Grand Total — Mid — Headline ($4,000 base x composite x tax)	—	$21,600
Grand Total — High ($4,500 base x composite x tax)	—	$24,400

7. COST CONVERSION SUMMARY

--- Final locked cost figures for this estimate. Identical to MANIFEST.

Field	Value
Fabricated Tonnage	4.92 t
Rate Basis	DEFAULT
Rate — Low	$3,500 per ton
Rate — Mid	$4,000 per ton
Rate — High	$4,500 per ton
Composite Adjustment Factor	x1.10
Adjusted Rate — Low	$3,850 per ton
Adjusted Rate — Mid	$4,400 per ton

Field	Value
Adjusted Rate — High	$4,950 per ton
Country and Tax	USA at 0%
Grand Total — Low (inc. tax)	$18,900
Grand Total — Mid — Headline (inc. tax)	$21,600
Grand Total — High (inc. tax)	$24,400
Cost per Fabricated Ton — Mid	$4,390 per ton

8. ASSUMPTIONS AND EXCLUSIONS

--- Assumptions applied in this estimate — minimum five items — cite sheets:

• Material grade for members taken as ASTM A992, A500, or A36 per General Structural Notes on sheet S0.01.

• Surface treatment taken as Shop primer only per General Structural Notes on sheet S0.01.

• Assembly complexity classified as Mixed bolt and weld based on connection types found on sheet S0.02.

• Composite factor of x1.10 is the product of x1.00 material, x1.00 surface, and x1.10 assembly factors.

• A 3.00% addition to base steel tonnage applied for bolts and weld accessories as a standard per-ton allowance.

• Rate band of $3,500 to $4,500 per fabricated ton applied based on current USA market rates for this member type and complexity.

• Member lengths are estimated from scaled dimensions on framing plan S1.02.

• Trellis column height estimated at 8'-0" based on typical section 2/S3.06.

Exclusions — not included in this estimate:

• Site erection labour, cranes, manlifts, and rigging.

• Engineer of Record stamping and delegated connection design.

• Loose anchor bolts cast-in-place by the general contractor.

• Site touch-up painting following erection.

• Tax or duties beyond 0% applied in this figure.

• 3D MEP coordination and cross-trade clash detection.

• Inflation beyond 30 calendar days from the date of this estimate.

• All steel lintels (W8x24, W16x50, Angles) and columns (C-marks) listed in schedules on S0.02, as quantities and locations are not defined on plans.

• All HSS members (HSS18x6x3/8, HSS9x5x3/8, HSS8x6x3/8) for which the profile is not listed in the provided AISC Unit Weight Reference table.

Conditions that produce a revised estimate:

• Tonnage change exceeding plus or minus 5% from this figure.

• Change of coating system such as from shop primer to hot-dip galvanised.

• Addition of member categories not in the current drawing set.

• Profile size changes affecting more than 10% of total tonnage.

10. ESTIMATE SUMMARY

--- The total fabricated tonnage for this estimate is 4.92 metric tons, with a headline cost of $21,600 USD based on the midpoint rate and including a composite adjustment factor of x1.10 and 0% tax. W-shape beams comprise the largest portion of the total tonnage. The estimate is based on the default rate band of $3,500 to $4,500 per fabricated ton. The scope of this estimate explicitly excludes all steel lintels and columns shown only in schedules, site erection labor, delegated connection design, and all HSS profiles not found in the provided unit weight table. A revised estimate will be produced if the total tonnage changes by more than 5% or if the specified coating system is altered.